



ECEN 314 - Signals and Systems

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Module 14 - Frequency Domain Analysis of Circuits/Systems II
Bode Plots

Learning Objectives



- Given a transfer function with poles and zeros at various frequencies, be able to sketch a straight line approximation to the magnitude response (Bode magnitude plot).
- Given a transfer function with poles and zeros at various frequencies, be able to sketch a straight line approximation to the phase response (Bode phase plot).

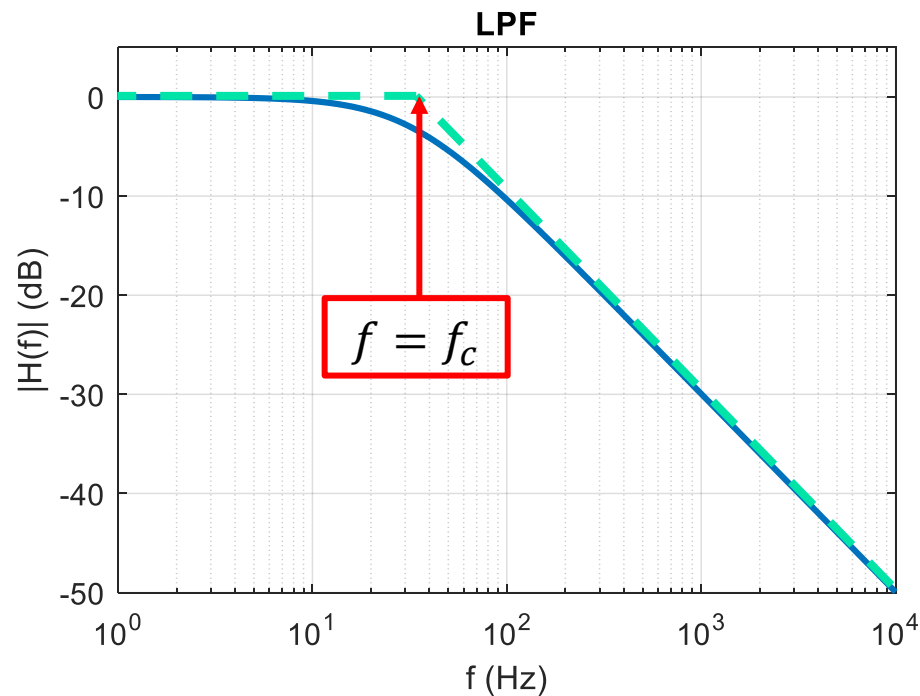
Bode Plots – Magnitude



Bode plots are approximations to the magnitude and phase response that we construct by hand using a series of linear approximations. Recall for our RC LPF example:

$$H(\omega) = \frac{1}{1+j\omega RC}, \quad |H(\omega)|^2 = \frac{1}{1+(\omega RC)^2}, \quad 10 \log_{10}(|H(\omega)|^2) = -10 \log_{10}(1 + (\omega RC)^2)$$

Define $\omega_c = \frac{1}{RC}$ so that $|H(\omega)|^2 = \frac{1}{1+(\frac{\omega}{\omega_c})^2}$ or $|H|_{dB} = -10 \log_{10}\left(1 + \left(\frac{\omega}{\omega_c}\right)^2\right)$.



For $\omega \ll \omega_c$, $|H|_{dB} \approx 0$,

- ✓ Slope = 0,
- ✓ Y-intercept = 0.

For $\omega \gg \omega_c$,

$$|H|_{dB} \approx -20 \log_{10}(\omega) + 20 \log_{10}(\omega_c)$$

- ✓ Slope = -20 (dB/decade)
- ✓ Y-intercept = $20 \log_{10}(\omega_c)$.

These two straight line approximations intersect at $\omega = \omega_c$.

Note: At $\omega = \omega_c$, the actual magnitude is $|H|_{dB} = -10 \log_{10}(2) = -3.01 \text{ dB}$

Bode Plots – Magnitude



Now suppose we have a LPF with two poles:

$$H(\omega) = \frac{1}{(1+j\frac{\omega}{\omega_1})(1+j\frac{\omega}{\omega_2})}$$

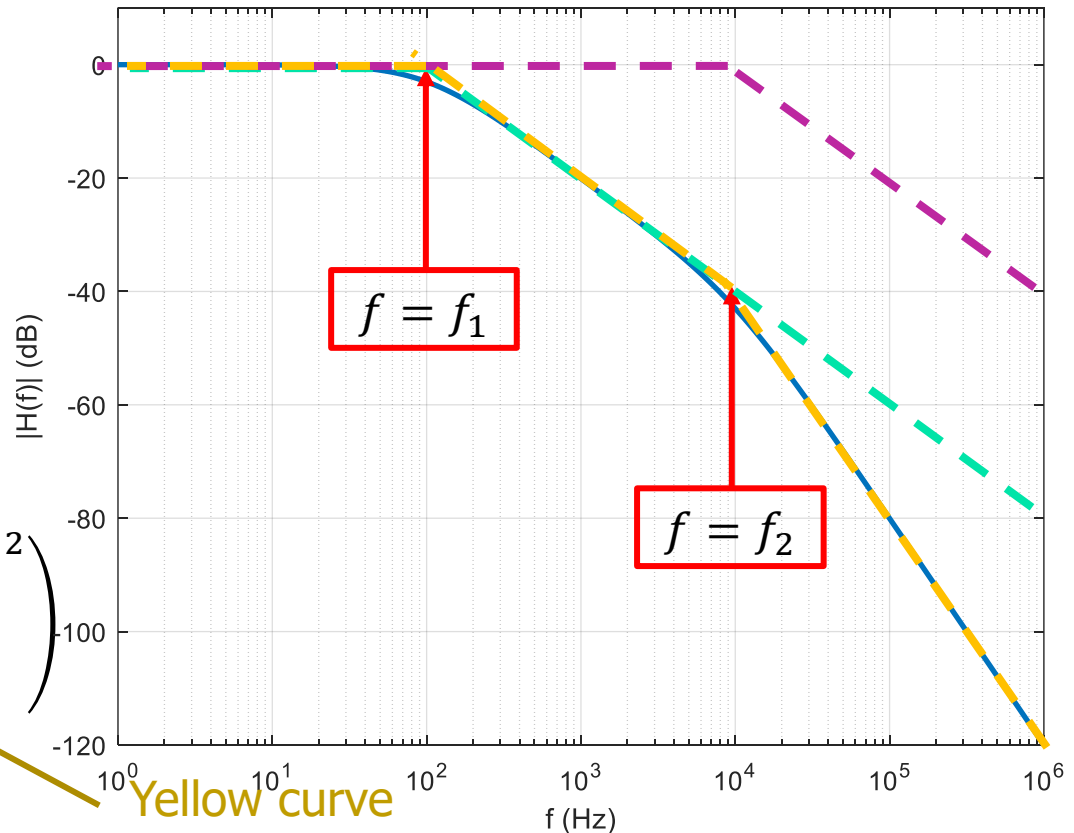
The magnitude response in dB is then

$$10 \log_{10}(|H(\omega)|^2)$$

$$= 10 \log_{10} \left(\left| \frac{1}{(1+j\frac{\omega}{\omega_1})(1+j\frac{\omega}{\omega_2})} \right|^2 \right)$$

$$= -10 \log_{10} \left(\left| 1 + j\frac{\omega}{\omega_1} \right|^2 \right)$$

$$- 10 \log_{10} \left(\left| 1 + j\frac{\omega}{\omega_2} \right|^2 \right)$$



Yellow curve

Green curve

Purple curve

Each term can be approximated like the last example.

Bode Plots – Magnitude

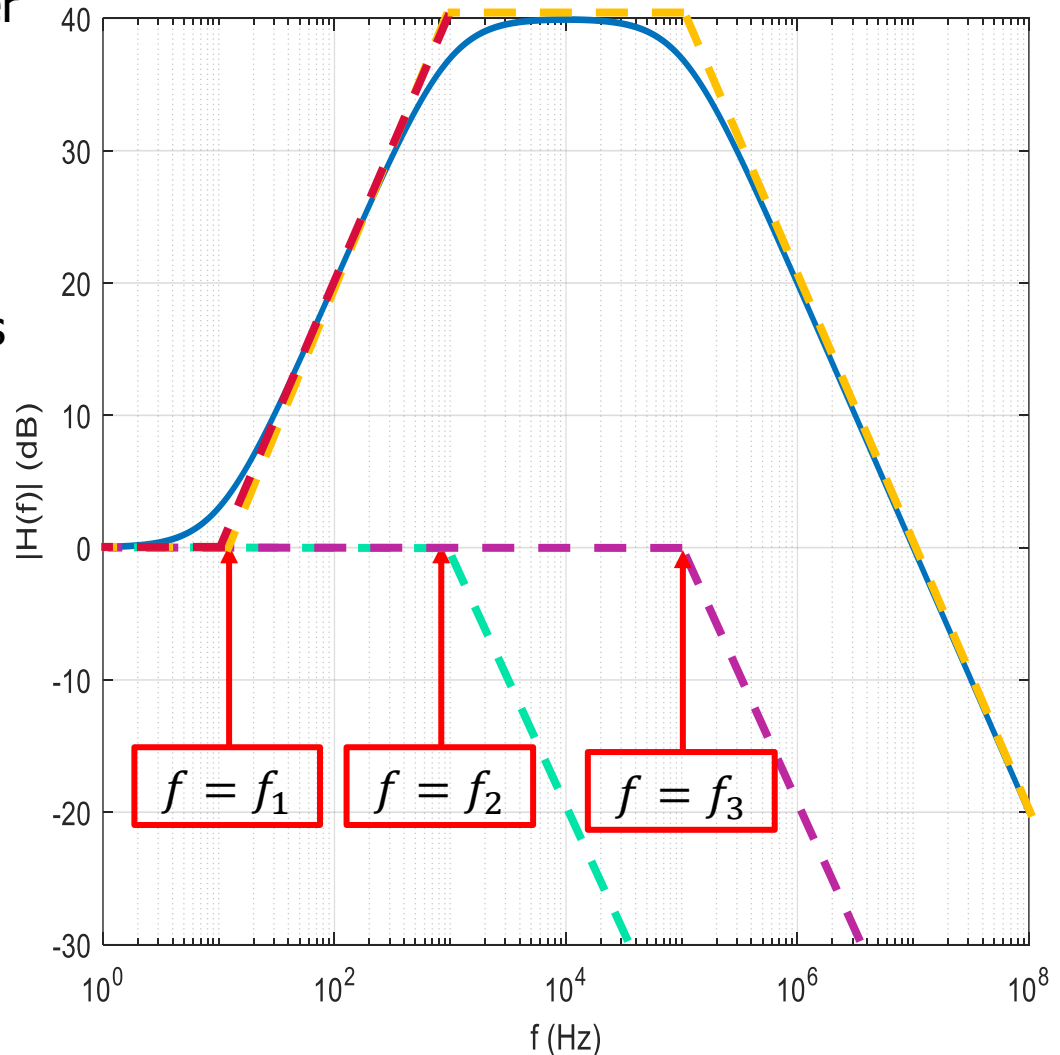


Zeros can be treated in a manner similar to poles:

$$H(\omega) = \frac{(1+j\frac{\omega}{\omega_1})}{(1+j\frac{\omega}{\omega_2})(1+j\frac{\omega}{\omega_3})}$$

The magnitude response in dB is then

$$\begin{aligned} 10 \log_{10}(|H(\omega)|^2) = & \\ & +10 \log_{10} \left(\left| 1 + j \frac{\omega}{\omega_1} \right|^2 \right) \\ & - 10 \log_{10} \left(\left| 1 + j \frac{\omega}{\omega_2} \right|^2 \right) \\ & - 10 \log_{10} \left(\left| 1 + j \frac{\omega}{\omega_3} \right|^2 \right) \end{aligned}$$



Bode Plots – Magnitude



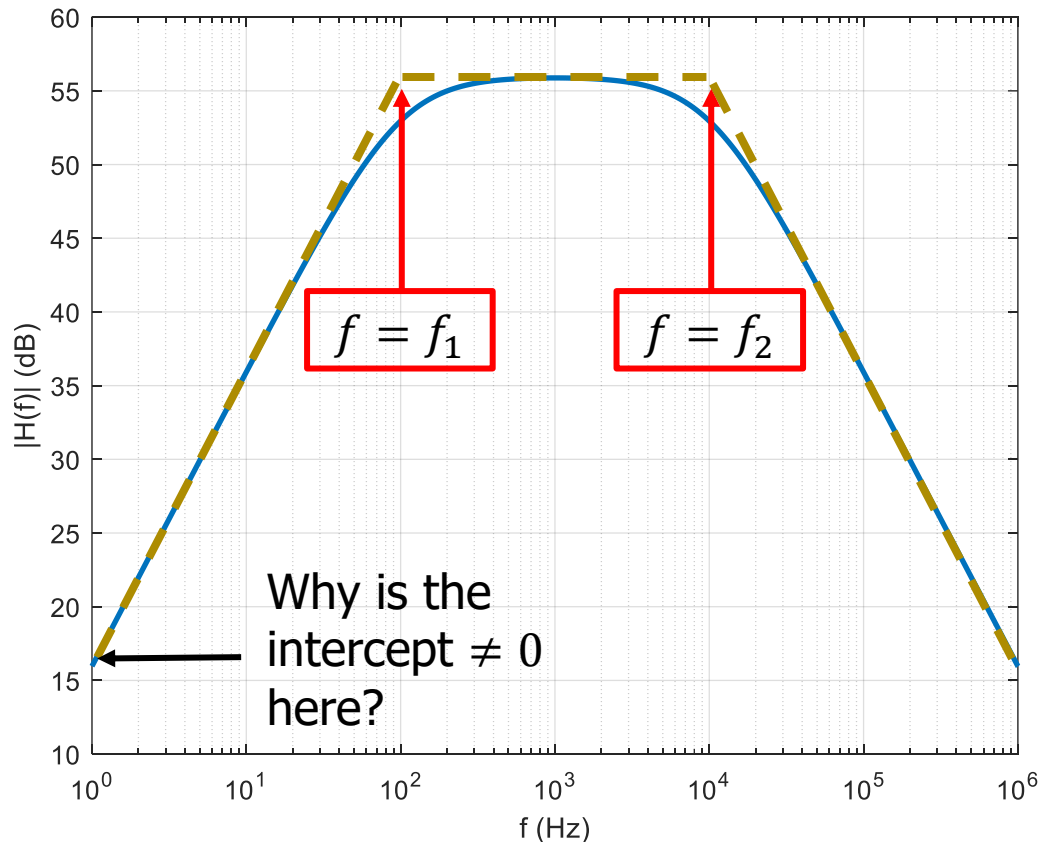
Poles or zeros at the origin can be dealt with exactly:

$$H(\omega) = \frac{j\omega}{\left(1+j\frac{\omega}{\omega_1}\right)\left(1+j\frac{\omega}{\omega_2}\right)}.$$

The magnitude response in dB is then

$$\begin{aligned} 10 \log_{10}(|H(\omega)|^2) = & \\ +20 \log_{10}(\omega) & \\ -10 \log_{10} \left(\left| 1 + j \frac{\omega}{\omega_1} \right|^2 \right) & \\ -10 \log_{10} \left(\left| 1 + j \frac{\omega}{\omega_2} \right|^2 \right) & \end{aligned}$$

The term due to the zero @ $\omega = 0$ has a slope of +20dB/decade and y-intercept of 0 and is exactly linear.



For $\omega_1 \ll \omega \ll \omega_2$ (the flat part), the magnitude response should be approximately $|H(\omega)| \approx \omega_1$
 $\rightarrow |H|_{dB} \approx 20 \log_{10}(\omega_1).$

Bode Plots – Magnitude



Second order underdamped systems will produce poles in complex-conjugate pairs. In which case the transfer function and magnitude response will look something like

$$H(\omega) = \frac{1}{\left(1 + \frac{j\omega}{\alpha + j\beta}\right)\left(1 + \frac{j\omega}{\alpha - j\beta}\right)} \rightarrow |H(\omega)|^2 = \frac{1}{\left(1 - \frac{\omega^2}{\alpha^2 + \beta^2}\right)^2 + \left(\frac{2\alpha\omega}{\alpha^2 + \beta^2}\right)^2}$$

Define $\omega_o = \sqrt{\alpha^2 + \beta^2} = |\alpha + j\beta|$. Then

✓ For $\omega \ll \omega_o$, $|H(\omega)|^2 \approx 1 \rightarrow |H|_{dB} \approx 0$,

✓ For $\omega \gg \omega_o$, $|H(\omega)|^2 \approx \frac{\omega_o^4}{\omega^4} \rightarrow |H|_{dB} \approx 40 \log_{10}(\omega_o) - 40 \log_{10}(\omega)$.

At very small and very large frequencies, the complex conjugate pair behaves similar to a double real pole at ω_o .

At $\omega = \omega_o$, the magnitude response will be:

$$|H(\omega_o)|^2 = \left(\frac{\omega_o}{2\alpha}\right)^2 \text{ or } |H|_{dB} = 20 \log_{10} \left(\frac{\omega_o}{2\alpha}\right)$$

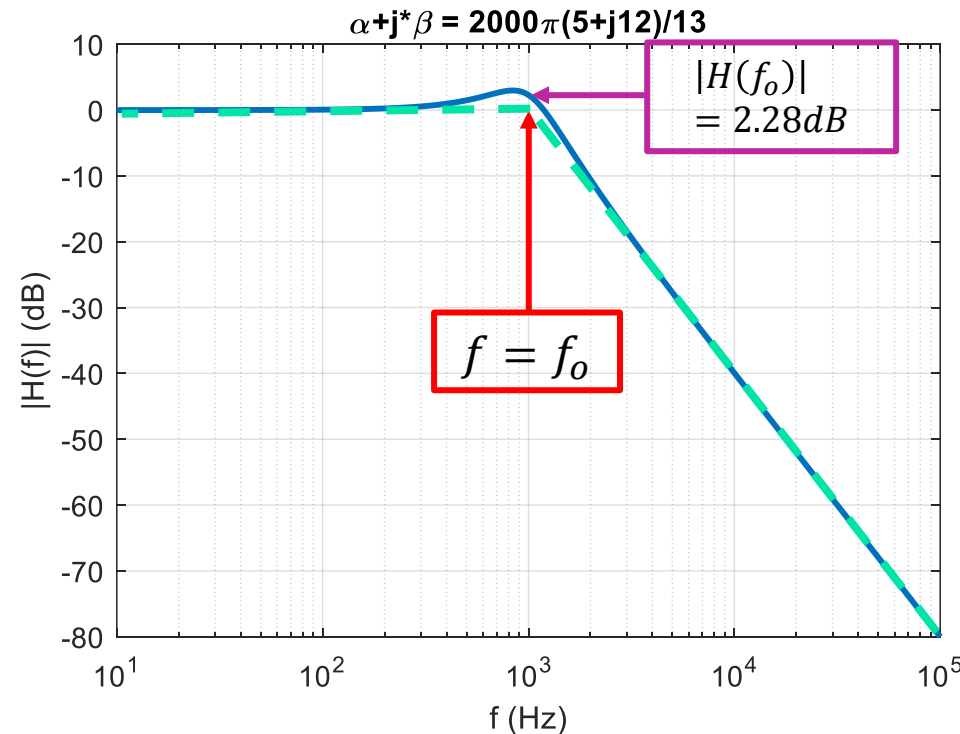
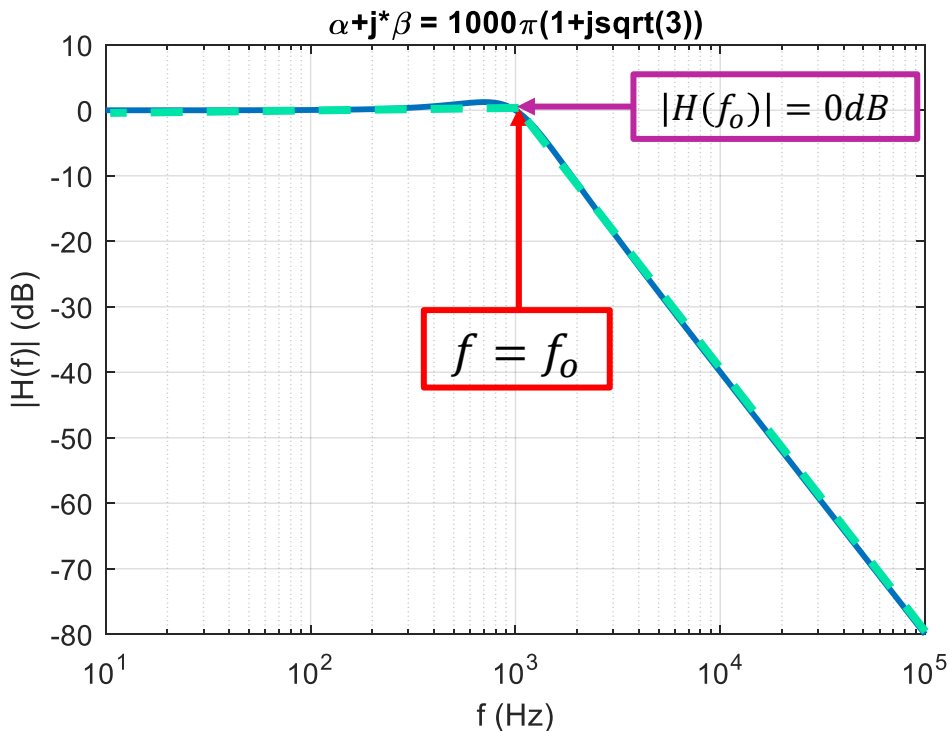
Note: value depends on $\alpha + j\beta$ and is not a fixed -3dB like the other cases.

Bode Plots – Magnitude



Two examples of underdamped second order LPFs ($f_o=1\text{kHz}$)

$$H(\omega) = \frac{1}{\left(1 + \frac{j\omega}{\alpha + j\beta}\right)\left(1 + \frac{j\omega}{\alpha - j\beta}\right)} \rightarrow |H(\omega)|^2 = \frac{1}{\left(1 - \frac{\omega^2}{\alpha^2 + \beta^2}\right)^2 + \left(\frac{2\alpha\omega}{\alpha^2 + \beta^2}\right)^2}$$



Example



Example 14.1: Sketch the Bode magnitude plot for the transfer function given by:

$$H(\omega) = \frac{\left(1 + j \frac{\omega}{\omega_2}\right) \left(1 + j \frac{\omega}{\omega_3}\right)}{\left(1 + j \frac{\omega}{\omega_1}\right) \left(1 + j \frac{\omega}{\omega_4}\right)}$$

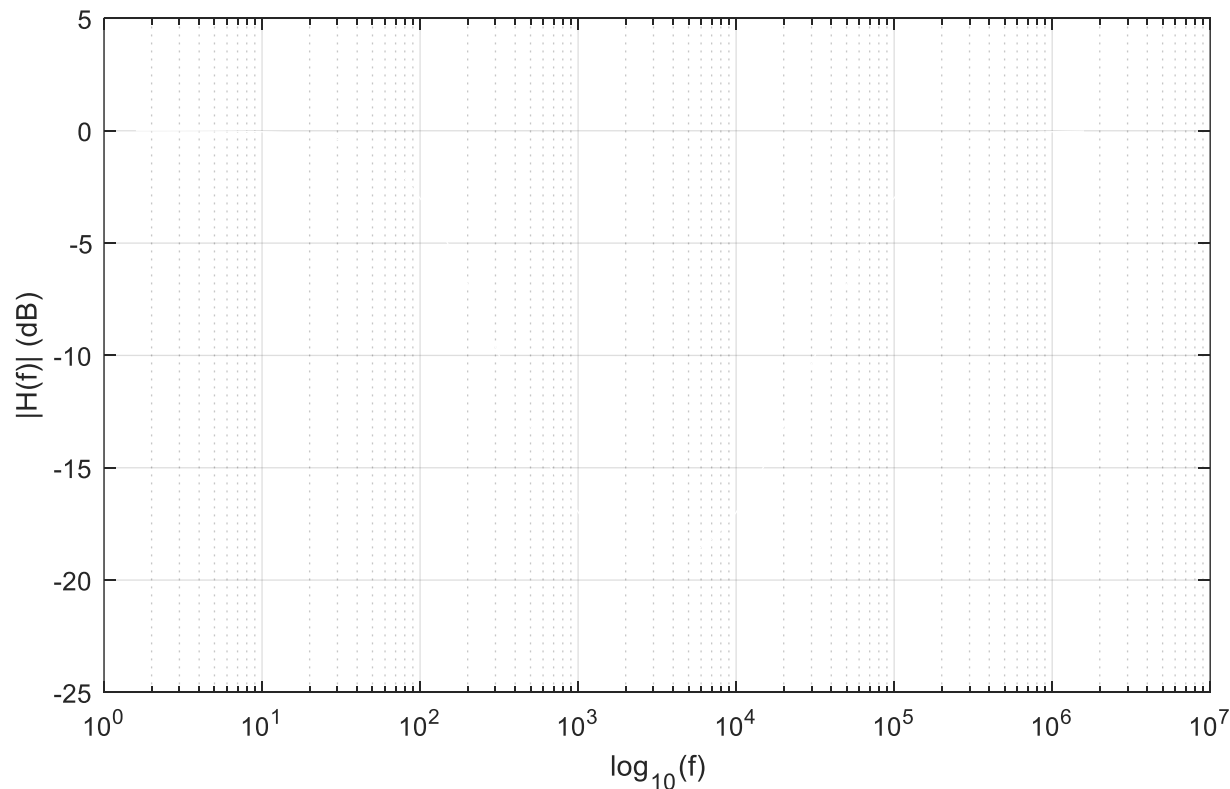
with

$$\omega_1 = 200\pi \text{ rad/sec,}$$

$$\omega_2 = 2000\pi \text{ rad/sec,}$$

$$\omega_3 = 20,000\pi \text{ rad/sec,}$$

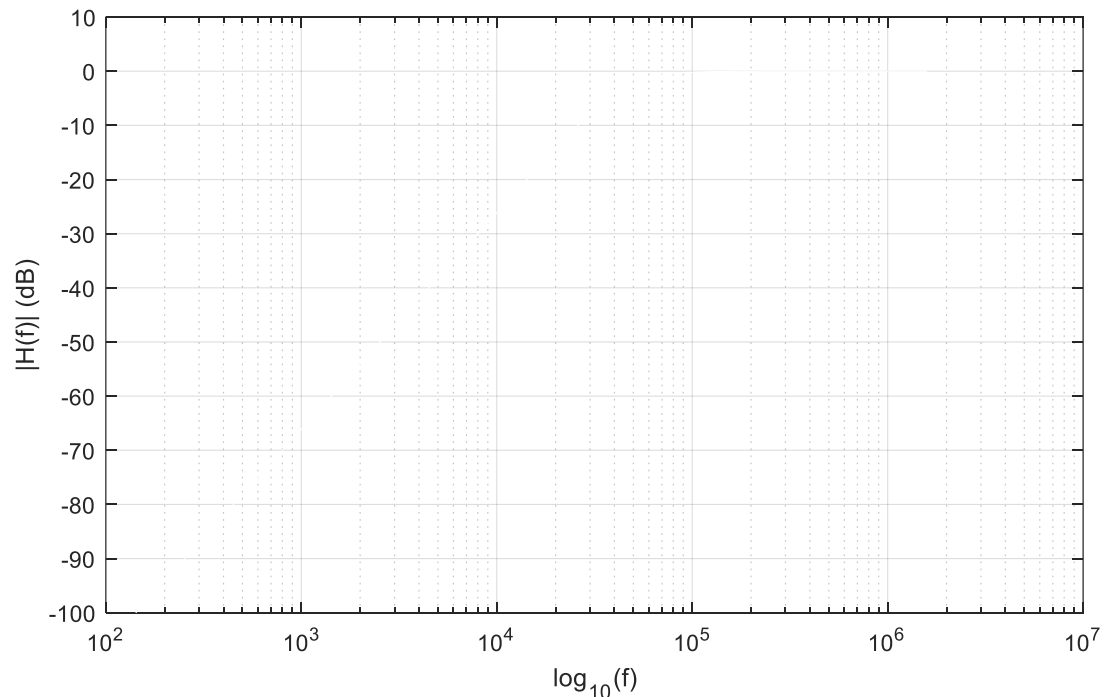
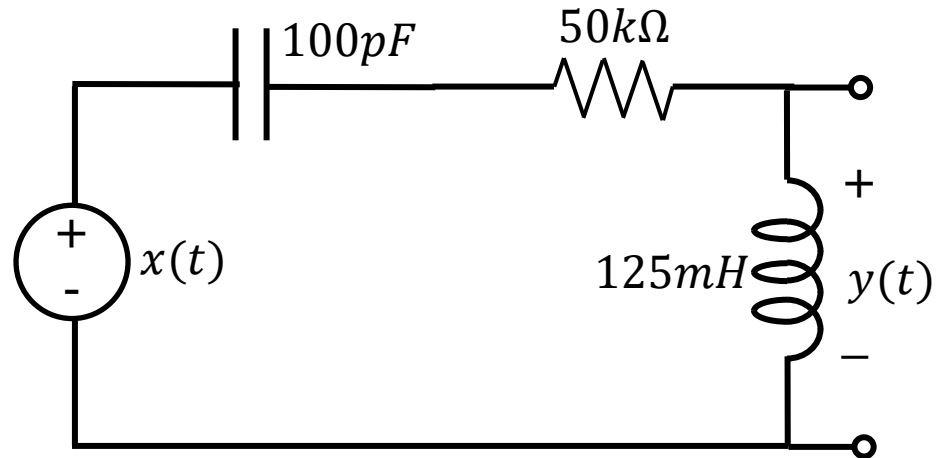
$$\omega_4 = 200,000\pi \text{ rad/sec.}$$



Example



Example 14.2: Sketch the Bode magnitude plot for the circuit shown.



Bode Plots – Phase



Approximations to the phase response can be formed in a manner similar to the magnitude response. Consider a single pole filter with a cutoff frequency at ω_c :

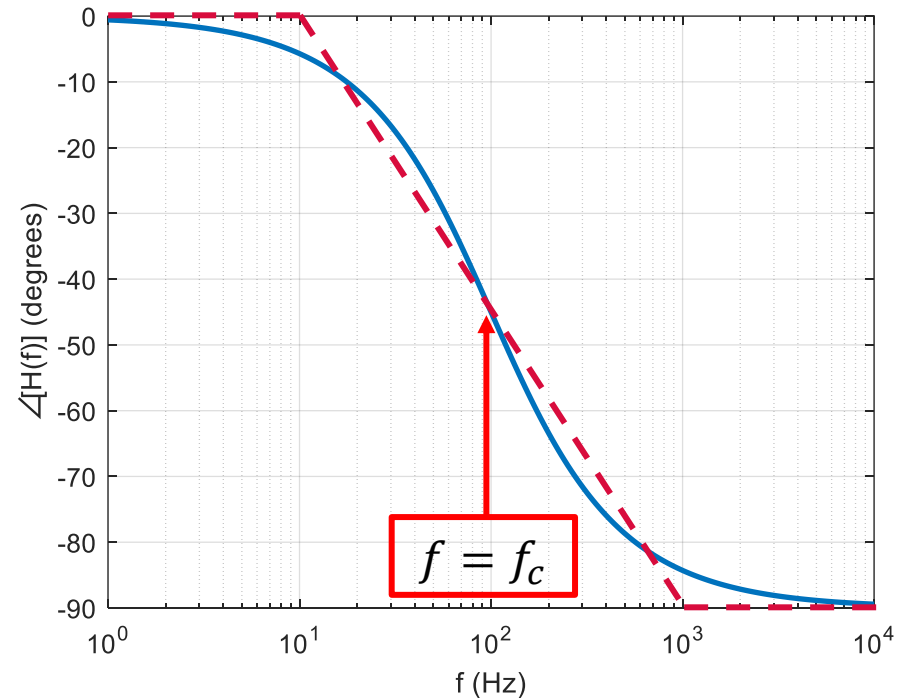
$$H(\omega) = \frac{1}{1+j\omega/\omega_c}$$

$$\angle H(\omega) = -\tan^{-1}(\omega/\omega_c).$$

This can be approximated by:

$$\angle H(\omega) = \begin{cases} 0^\circ, & \omega < \omega_c/10, \\ -45^\circ - 45 \log_{10}(\omega/\omega_c)^\circ, & \frac{\omega_c}{10} < \omega < 10\omega_c \\ -90^\circ, & \omega > 10\omega_c. \end{cases}$$

A zero would have the mirror image (flipped vertically) phase response. It would start at 0° , end at $+90^\circ$, and pass through $+45^\circ$ at ω_c .



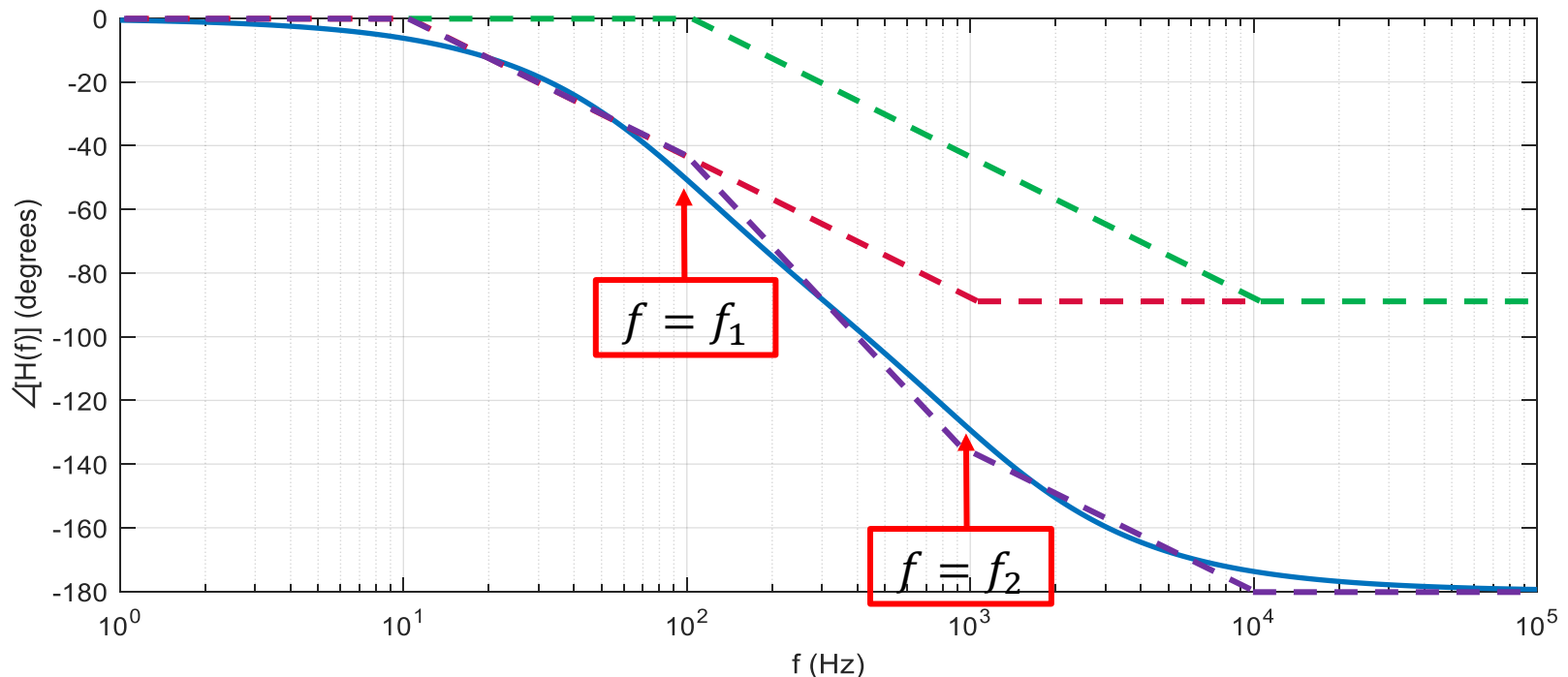
Bode Plots – Phase



If there are multiple poles/zeros, the contributions from each term can be added together:

$$H(\omega) = \frac{1}{\left(1 + \frac{j\omega}{\omega_1}\right)\left(1 + \frac{j\omega}{\omega_2}\right)}$$

$$\angle H(\omega) = -\tan^{-1}\left(\frac{\omega}{\omega_1}\right) - \tan^{-1}\left(\frac{\omega}{\omega_2}\right).$$



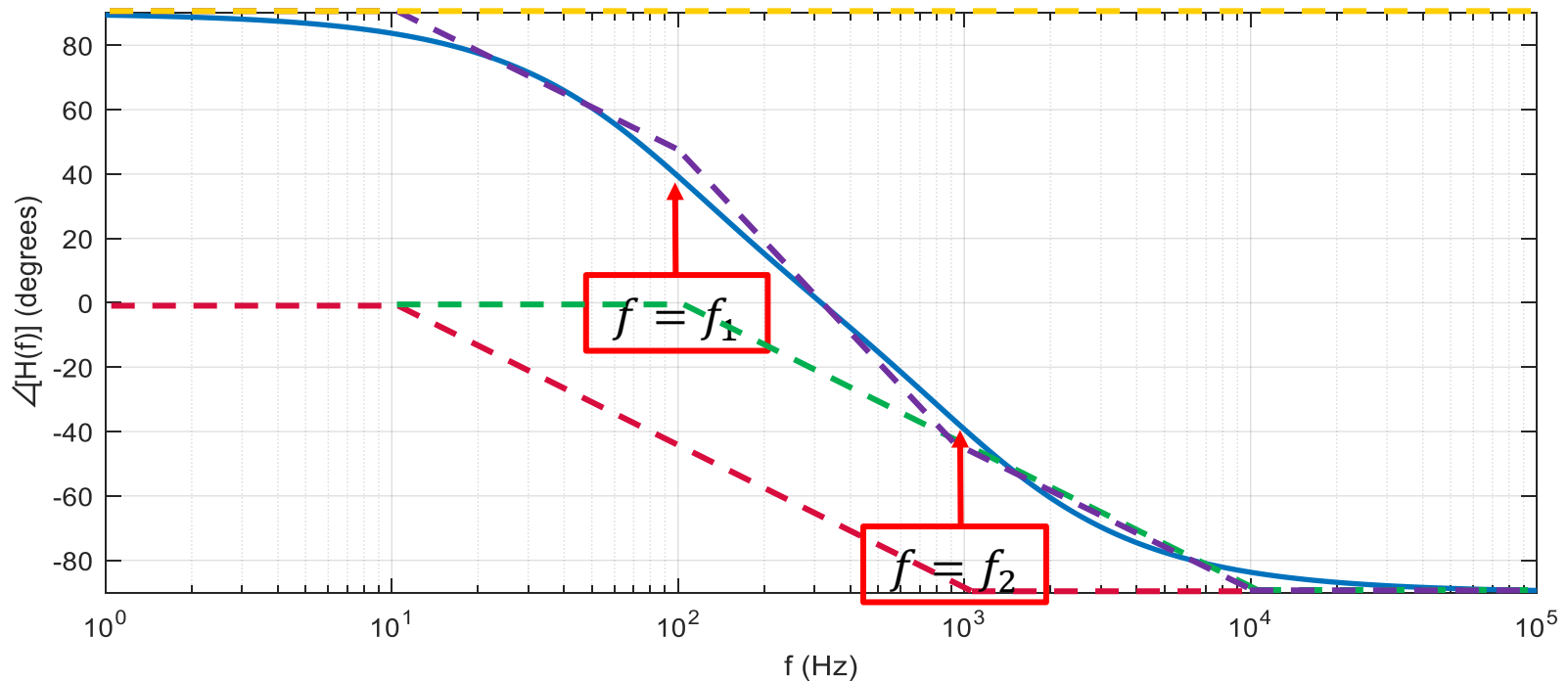
Bode Plots – Phase



A pole (or zero) at the origin will shift the phase response down (or up) by $+90^\circ$.

$$H(\omega) = \frac{j\omega}{\left(1 + \frac{j\omega}{\omega_1}\right)\left(1 + \frac{j\omega}{\omega_2}\right)'}$$

$$\angle H(\omega) = \frac{\pi}{2} - \tan^{-1}\left(\frac{\omega}{\omega_1}\right) - \tan^{-1}\left(\frac{\omega}{\omega_2}\right).$$

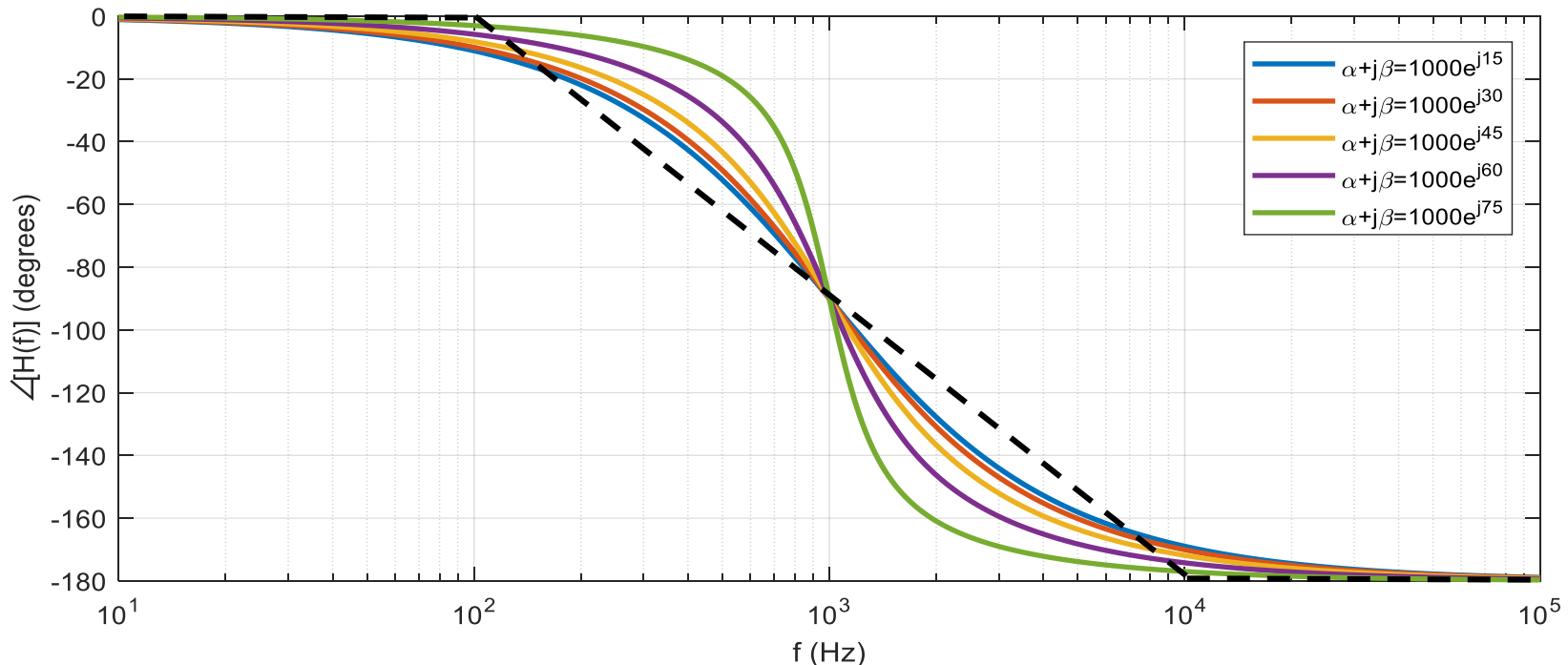


Bode Plots – Phase



A pair of complex conjugate poles (or zeros) can be crudely treated like a double pole/zero, but the straight-line approximation may be quite different than the real phase response.

$$H(\omega) = \frac{1}{\left(1 + \frac{j\omega}{\alpha + j\beta}\right)\left(1 + \frac{j\omega}{\alpha - j\beta}\right)}$$



Example



Example 14.3: Sketch the Bode phase plot for the transfer function given by:

$$H(\omega) = \frac{\left(1 + j \frac{\omega}{\omega_2}\right) \left(1 + j \frac{\omega}{\omega_3}\right)}{\left(1 + j \frac{\omega}{\omega_1}\right) \left(1 + j \frac{\omega}{\omega_4}\right)}$$

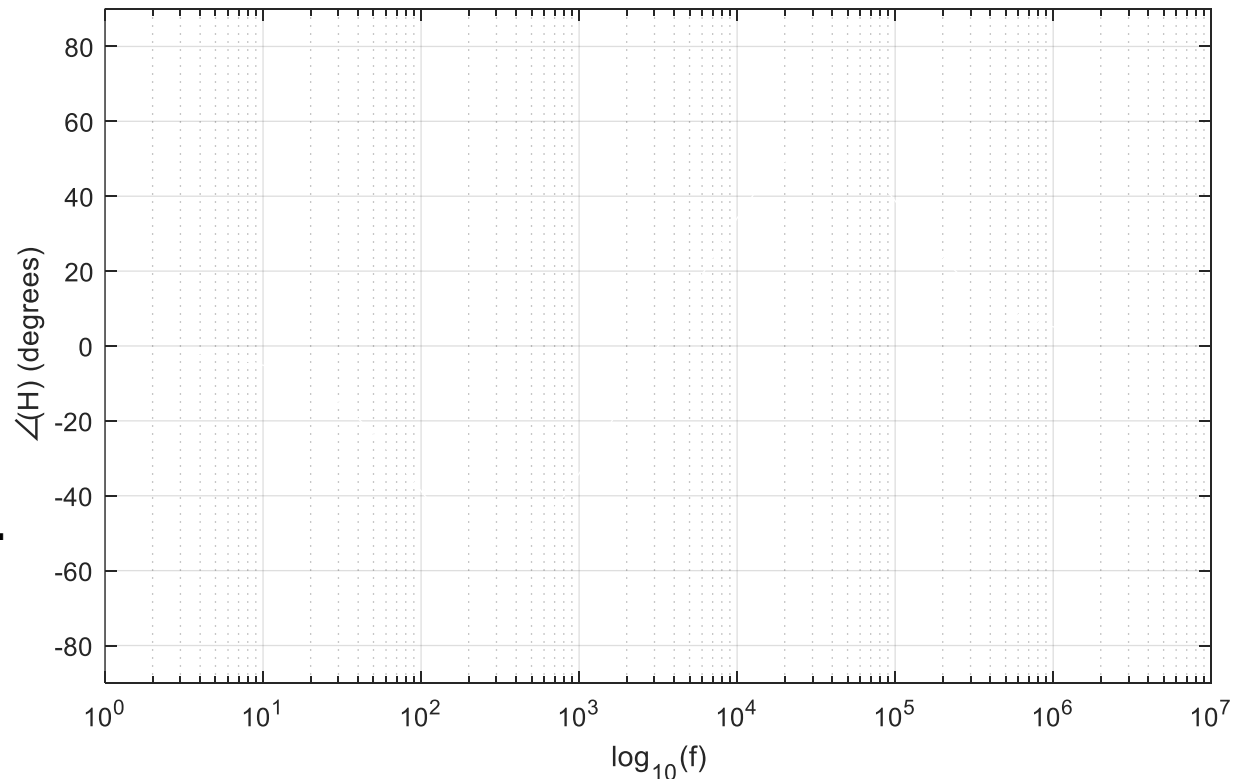
with

$$\omega_1 = 200\pi \text{ rad/sec,}$$

$$\omega_2 = 2000\pi \text{ rad/sec,}$$

$$\omega_3 = 20,000\pi \text{ rad/sec,}$$

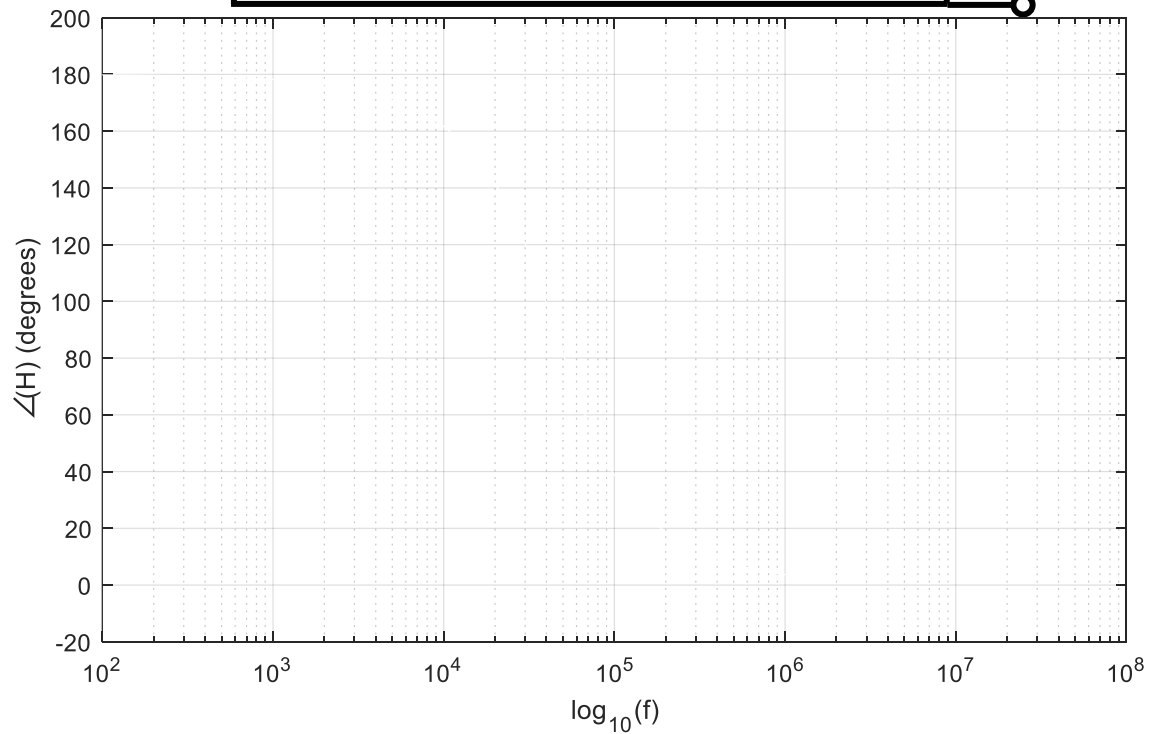
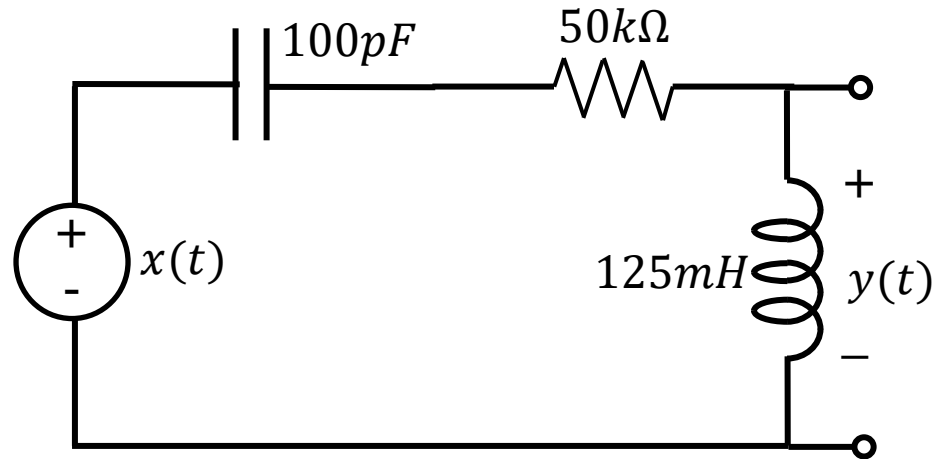
$$\omega_4 = 200,000\pi \text{ rad/sec.}$$



Example



Sketch the Bode phase plot for the circuit shown.



Summary of Rules for Bode Plots



Term	Magnitude	Phase
Constant K	$20 \log_{10}(K)$	0° if $K > 0$ $\pm 180^\circ$ if $K < 0$
Real pole $\frac{1}{1 + j\omega/\omega_c}$	Low freq @ 0 High freq slope = -20dB/dec Intersect at $\omega = \omega_c$	Low freq @ 0° High freq @ -90° Connect between $\omega_c/10 < \omega < 10\omega_c$
Real zero $1 + j\omega/\omega_c$	Low freq @ 0 High freq slope = +20dB/dec Intersect at $\omega = \omega_c$	Low freq @ 0° High freq @ $+90^\circ$ Connect between $\omega_c/10 < \omega < 10\omega_c$
Pole @ origin $1/j\omega$	Slope = -20dB/dec passing through 0dB @ $\omega = 1$	-90°
Zero @ origin $j\omega$	Slope = +20dB/dec passing through 0dB @ $\omega = 1$	$+90^\circ$
Complex conjugate poles $\frac{1}{(1 + \frac{j\omega}{\alpha + j\beta})(1 + \frac{j\omega}{\alpha - j\beta})}$	Treat like double real pole @ $\omega_c = \sqrt{\alpha^2 + \beta^2}$	
Complex conjugate zeros $(1 + \frac{j\omega}{\alpha + j\beta})(1 + \frac{j\omega}{\alpha - j\beta})$	Treat like double real zero @ $\omega_c = \sqrt{\alpha^2 + \beta^2}$	